# Project Design Phase-II
# Data Flow Diagram & User Stories

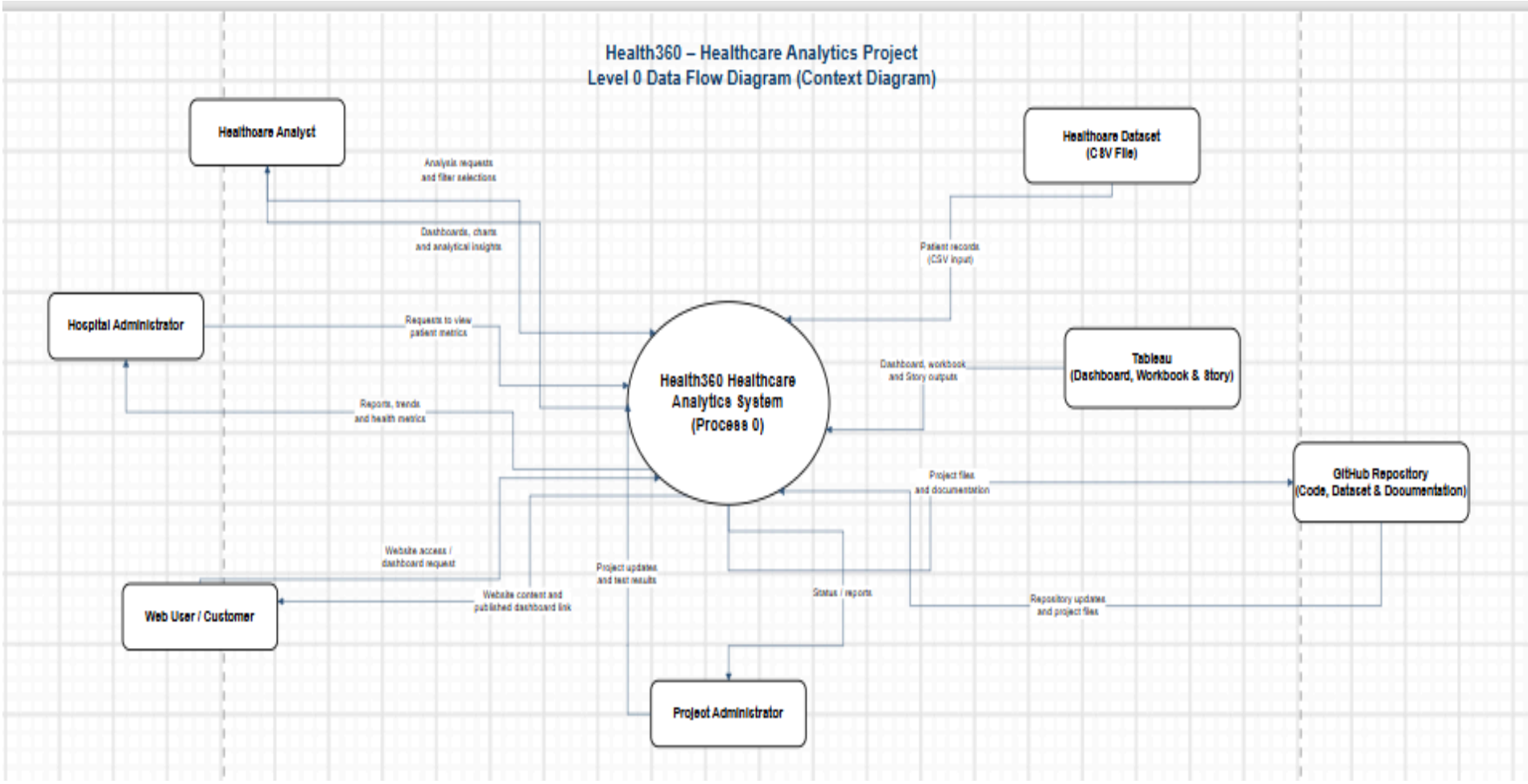
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|---|---|
| Date | 28 October 2026 |
| Team ID | EN25CS3090269 |
| Project Name | Smart Healthcare Analytics |
| Maximum Marks | 4 Marks |

**Data Flow Diagrams:**

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

**DFD Symbol Legend**

| Symbol | Name | Description |
|---|---|---|
| **Oval / Rounded shape** | **External Entity** | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
| **Rectangle with numbered header** | **Process** | An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order). |
| **Rectangle (solid fill, no number)** | **Data Store** | A place where data is held for later use (e.g., Customer Database, Inventory). |
| **Labeled arrow** | **Data Flow** | The movement of data between entities, processes, and data stores, labeled with what data is being passed. |


Health360 – Healthcare Analytics Project
Level 0 Data Flow Diagram (Context Diagram)
Healthcare Analyst
Healthcare Dataset
(CSV File)
Analysis requests
and filter selections
Dashboards, charts
and analytical insights
Patient records
(CSV input)
Hospital Administrator
Requests to view
patient metrics
Health360 Healthcare
Analytics System
(Process 0)
Dashboard, workbook
and Story outputs
Tableau
(Dashboard, Workbook & Story)
Reports, trends
and health metrics
Project files
and documentation
GitHub Repository
(Code, Dataset & Documentation)
Website access /
dashboard request
Project updates
and test results
Status / reports
Repository updates
and project files
Website content and
published dashboard link
Web User / Customer
Project Administrator

## User Stories

Use the below template to list all the user stories for the product.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Healthcare Analyst | Data Access & Integration | USN-1 | As a healthcare analyst, I want to access the healthcare dataset in CSV format so that I can analyze patient information. | The CSV dataset opens and connects to Tableau successfully. | High | Sprint-1 |
| Healthcare Analyst | Data Access & Integration | USN-2 | As a healthcare analyst, I want to connect the dataset to Tableau Desktop so that I can create healthcare visualizations. | Tableau loads the required fields with appropriate data types. | High | Sprint-1 |
| Healthcare Analyst | Data Processing | USN-3 | As a healthcare analyst, I want to prepare and validate the dataset so that the analysis is accurate. | Missing values, data types, and invalid values are checked and handled appropriately. | High | Sprint-1 |
| Healthcare Analyst | Data Processing | USN-4 | As a healthcare analyst, I want to categorize patients by age group and BMI category so that I can compare different patient groups. | Age groups and BMI categories are calculated and displayed correctly. | High | Sprint-1 |
| Healthcare Analyst | Business Analysis | USN-5 | As a healthcare analyst, I want to view the total number of patients so that I can understand the size of the dataset. | The dashboard displays the correct patient count. | High | Sprint-1 |
| Healthcare Analyst | Business Analysis | USN-6 | As a healthcare analyst, I want to analyze diabetes, heart disease, and stroke cases so that I can identify disease patterns. | The dashboard displays correct case counts based on the disease indicators. | High | Sprint-1 |
| Healthcare Analyst | Business Analysis | USN-7 | As a healthcare analyst, I want to compare average blood pressure across age groups so that I can identify variations between groups. | The chart displays average blood pressure for each age group. | High | Sprint-2 |
| Healthcare Analyst | Business Insights & Analytical Outputs | USN-8 | As a healthcare analyst, I want to view charts showing patient distribution by age group so that I can understand the patient demographics. | The age-group chart displays the correct patient distribution. | High | Sprint-1 |
| Healthcare Analyst | Business Insights & Analytical Outputs | USN-9 | As a healthcare analyst, I want to analyze heart disease by BMI category and exercise level so that I can explore relationships between these factors. | The heatmap displays the relevant BMI and exercise-level categories with corresponding values. | High | Sprint-2 |
| Healthcare Analyst | User Interaction & Analysis | USN-10 | As a healthcare analyst, I want to filter the dashboard by age group so that I can focus on specific age ranges. | Selecting an age group updates the relevant visualizations. | High | Sprint-2 |

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Healthcare Analyst | User Interaction & Analysis | USN-11 | As a healthcare analyst, I want to filter the dashboard by gender and BMI category so that I can compare patient groups. | The gender and BMI filters update the relevant charts correctly. | High | Sprint-2 |
| Healthcare Analyst | Data Update & Reporting | USN-12 | As a healthcare analyst, I want to refresh the dataset when updated data is available so that the dashboard reflects the latest available information. | After the data source is updated and refreshed, the workbook displays the updated values. | Medium | Sprint-3 |
| Hospital Administrator | Business Insights & Analytical Outputs | USN-13 | As a hospital administrator, I want to view the healthcare dashboard so that I can understand key patient statistics and trends. | The published dashboard displays KPI cards and charts clearly. | High | Sprint-2 |
| Hospital Administrator | Data Update & Reporting | USN-14 | As a hospital administrator, I want to view the healthcare analysis Story so that I can understand the project findings step by step. | All published Story points open and display their intended visualizations. | Medium | Sprint-3 |
| Customer (Web User) | Website Access | USN-15 | As a web user, I want to open the Health360 website so that I can learn about the project and its healthcare analytics features. | The website loads successfully and displays the relevant project information. | High | Sprint-2 |
| Customer (Web User) | Website Navigation | USN-16 | As a web user, I want to navigate between the website pages so that I can find the information I need. | The available navigation links open the correct pages. | Medium | Sprint-2 |
| Customer (Web User) | Dashboard Access | USN-17 | As a web user, I want to access the published Tableau dashboard from the website so that I can explore the healthcare analysis. | The dashboard link opens the correct published Tableau dashboard. | High | Sprint-3 |
| Project Administrator | Project Management | USN-18 | As a project administrator, I want to maintain the project files and documentation in GitHub so that the source code, dataset, and workbook are organized. | The repository contains the relevant project files and a clear README. | High | Sprint-1 |
| Project Administrator | Reliability & Testing | USN-19 | As a project administrator, I want to test the dashboard filters, charts, Story, and website so that I can identify errors before submission. | All listed features are tested and any identified errors are documented. | High | Sprint-3 |
| Project Administrator | Security | USN-20 | As a project administrator, I want to ensure that public project files do not expose confidential patient information so that the project can be shared safely. | Only synthetic or appropriately anonymized data is shared publicly. | High | Sprint-1 |